Natural Class*

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Abstract

Which of these are natural classes?:

- {t, d}
- {t}
- {d}
- {t, D, d}
- {D}
- {t, D}
- {D, d}

The answers fall out from basic logic, perhaps best understood as "third factor" (Chomsky 2005) effects.

1 Introduction

The notion of natural classes is ubiquitous in the phonology literature, as well as in other domains of linguistics. Despite the fundamental nature of the notion, there are many inconsistencies, misconceptions and logical errors in discussion of natural classes. In this paper I identify some of the sources of these problems and offer an explicit definition of the notion in the domain of classes (or sets) of individual phonological segments. Larger units, such as strings, syllables, morphemes, and so on, will not be handled here, although all can be combined into a general discussion of the distinction between individual representations and their relationship to the "partial descriptions" or equivalence classes that intensionally characterize natural classes of such representations. Given the concerns of this paper, I treat segments as *sets* that contain feature specifications as their (unordered) elements, abstracting away from issues like potential feature geometric structures or other hierarchical organization of phonological primitives (e.g., Sagey 1986).

*Thanks to Kyle Gorman and Fae Hicks for useful suggestions.

In order to ground the discussion with concrete examples, consider a language with just two coronal stops, /t/ and /d/: does the set {t, d} constitute a natural class? Does a set with just one member, say, {t}, constitute a natural class? Consider now a language with not only /t/ and /d/, but also /D/, a coronal stop unspecified for voicing. Does the set {t, D, d} constitute a natural class? How about the sets {t, D} and {D, d}? Finally, what about the set {D}? The answers, and some of the notions presupposed in these questions, obviously rely on the nature of the feature system adopted. Only after we lay out a particular system and a precise notion of natural class will we be able to insightfully answer these questions.

2 Feature systems and natural classes

A first point of clarification is that the set of natural classes definable within a system is dependent upon the set of available representational primitives. For example, a privative feature system that contains, say, Consonantal and Voiced, but not Voiceless can make reference to all voiced consonants (those that contain the specification Voiced), or to all consonants, but there is no way to make reference to just the voiceless consonants. This claim, of course, holds only if the system provides no way to indicate "not Voiced." Obviously, a privative feature system that contains Consonantal and Voiceless, but not Voiced can make reference to all voiceless consonants, or all consonants, but not to just the voiced ones. Finally, a privative feature system that happens to have Consonantal, as well as both Voiced and Voiceless, can make reference to three classes, the class of all consonants, the class of voiced consonants and the class of voiceless consonants. This last situation, at first blush, mimics a binary system containing two valued features, +Voiced and −Voiced, but more needs to be said.

The privative system that happens to contain the two features, Voiced and Voiceless, clearly allows, by the nature of privativity, for segments to be lacking both of these features. In a binary system, the possibility of underspecification, absence of any value for, say Voiced, is, by fiat of the analyst, either allowed or disallowed. The system I explore below allows for underspecification. In either case, privativity with "opposite" features like Voiced and Voiceless, or binarity with underspecification, the question arises of whether the segments that lack a specification for voicing, like /D/ constitute by themselves a natural class.

Other feature systems are of course logically possible, and have been proposed—perhaps vowel height can be handled by a scalar system of three (or more) values in which low vowels contain Height 1, mid vowels contain Height 2 and high vowels contain Height 3 (see Clements 2015 who attributes the idea to Trubetzkoy 1939). If we adopt such a system, we must decide whether we build into the system the capacity to refer to "not Height 1," as a way to treat the mid and high vowels as a class, and this leads to the question of whether we can refer to "not Height 2" to treat the non-mid vowels as a class. Another possibility might be to allow reference to continuous portions of the scale, such as "Height $\geq$ 2."

This brief sketch of different feature systems raises the question of whether all segments in all languages are built from a single universal set of features, or whether learners of each language "construct features from scratch" (Householder 1965) and create their feature system, perhaps somehow finding "the best" system for the target language. In the remainder of this paper, I adopt the position of Chomsky and Halle (1965, 1968) and Reiss and Volenec (2022) that there is one universal (innate) feature set for spoken language: "As has often been noted [references omitted–cr] the assumption of a universal feature structure is made (often only implicitly) in every approach to phonology that is known, and clearly cannot be avoided. What is at issue only is the choice of features, not their universality" (Chomsky and Halle 1965:fn. 24). Note that this position also characterizes classic Optimality Theory work that assumes a universally defined constraint inventory (e.g., McCarthy and Prince 1993). I also assume that all features are binary—each can occur in a segment with either a + or − coefficient; and that underspecification is possible—it may be the case that a particular segment contains neither +F nor −F for some feature—because the model has no stipulation against it.[1]

Because I commit to a precise model of features, I will be able to pose and answer the precise questions about segments such as /t, D, d/ mentioned above, as well as more general questions of the following sort:

(1) Questions about natural classes:

- Given a universal set of *n* features, how many segments can be defined?
- Given a set of *n* features, how many *sets* (or classes) of segments can be defined?
- How can the traditional notion of natural class be formalized?
- Given a set of *n* features, how many sets of segments correspond to the traditional notion of natural class?
- How does the possibility of underspecification bear on the nature of natural classes.

3 Natural classes and rules

3.1 Rules are the "Laws" of a Grammar

Fodor and Piattelli-Palmarini (2010:136) observe that "Nothing has a nomological explanation unless it belongs to a natural kind." This means that analyzing anything as

[1] I follow Inkelas in *not* treating the goal of minimization of information in the lexicon as a justification for underspecification: LP is not concerned with the lexical encoding of so-called redundant features. The idea that allowing underspecification with binary features is somehow an illicit way to 'sneak in' a third value can be addressed by pointing out the existence of so-called three-valued constructivist logics, accepted by logicians and theoretical computer scientists (e.g., 'How many is two?' at `https://math.andrej.com/2005/05/16/how-many-is-two/`]. In this paper, the possibility of underspecification is an undefended postulate which allows us to explore how underspecification interacts with natural class logic.

lawful in any domain is only possible if the entities involved are defined in terms of classes of elements relevant to the laws of the domain. I take the phonological rules of a given language as parallel to Fodor and Piatelli-Palmarini's laws. I treat rules as functions mapping sets of strings (or more complex structures) of segments to other strings (or structures).[2] Fodor and Piatelli-Palmarini consider their statement to be a truism, basically, a definition of what it means to be a law. I argue that we need the same approach to phonological rules—so, we need to define the "classes of elements" to which phonological rules apply. Not surprisingly, this means that we need a precise notion of natural classes of segments (and other representational elements). I demonstrate below that this precise notion is lacking to various degrees in the literature, some of which implicitly or explicitly rejects the approach advocated by Fodor and Piatelli-Palmarini.

3.2 No rules without natural classes

The Logical Phonology framework (LP; e.g., Bale and Reiss 2018, Volenec and Reiss 2020, Reiss 2012, Gorman and Reiss 2025a, Gorman to appear) adopted here assumes that the syntax of rules is expressed in terms of natural classes in the targets of rules and in the environments as well. In other words, a formalization of the notion natural class is part of the formalization of what is a possible rule. This is implicit in much work, and occasionally stated directly. For example, Odden (2013:159) expresses the "central claim" that rule targets and environments *are* natural classes defined intensionally via "conjunctions of properties," and Spencer (1996:135) points out that the "concept of a natural class allows us to sharpen up what we mean by a phonological rule."[3]

However, the literature is rife with statements that indicate a lack of appreciation for the intimate relation between natural class logic and rule syntax. Hayes (2009) says that in rules, natural classes occur "in most instances" (p. 43) or as a "pervasive tendency" (p. 71). Kornai (2008), assuming a language with a segment inventory that includes at least [p, t, b, d], says that "Phonologists would be truly astonished to find a language where some rule or regularity affects *p*, *t*, and *d* but no other segment"—presumably he has *b* in mind. Mielke (2008:9) claims that "unnatural classes are widespread" in rules. The ideas is that many rules do *not* refer to natural classes—but these claim are empty since they are not accompanied by a theory of what is a possible rule. It appears to be the case that Mielke considers a rule to be at least anything that has been called a rule in the literature, but as we have seen,

[2]Two clarifications are in order: first, the parallel I am making is between scientific laws and rules of individual I-languages, not between the universal laws of phonology, if such things exist. Second, phonological rules are often expressed as functions mapping segments to segments in a given environment, but, for reasons discussed by (Bale and Reiss 2018:251-3), the segment-to-segment mapping view is problematic. A function f can compose with a function g (i.e.; $f \circ g$), if and only if the range of g is a subset of the domain of f. In other words, we need g's outputs to be possible inputs to f. If a rule literally did not apply because a particular segment was not in the input string, then the rule would have no output. So then there would be no input for the next rule. In our view, rules are functions mapping strings to strings, so an identity mapping of $xyz \rightarrow xyz$ by a rule provides an input for the next rule.

[3]The issue of *distinctness*, which plays a role in Chapter 8 of *SPE*, as well as in discussions such as Lightner (1963) and Harms (1968) can be safely ignored, given my acceptance of underspecification and lack of concern with redundancy in the lexicon.

one's natural classes depend on one's assumed feature system, and the literature is not consistent in the system adopted in each analysis.

Of course, it may be the case that a rule *affects* a set of segments that do not constitute a natural class, and thus *appears* to target a non-natural class. For example, if a language with /p, b, m/ has a rule deleting /b/ word-finally, followed by a rule palatalizing word-final labials, the second rule will appear to apply non-vacuously only to the non-natural class {p, m}. This is of course just a reflection of the bleeding rule ordering. The intensionally defined target of the palatalization rule will be the class of all labials. Using a classic example, Volenec and Reiss (2020) show that the natural classes that play a role in the phonological behavior of the English noun plural ending show exactly this logical structure, drawing a parallel between its acquisition and masking effects in visual and auditory perception. Stem-final voiceless obstruents devoice the underlying /-z/ of the plural suffix, so we get *cat*[s] but *dog*[z]. However, the voiceless obstruents that happen to also be coronal stridents fail to cause devoicing, not because the devoicing rule doesn't respect natural class logic, but because the rule that inserts a vowel in *mass*[-ɨz], *bush*[-ɨz] and *match*[-ɨz] precedes and bleeds the devoicing rule. The epenthesis rule "masks" some members (the voiceless coronal stridents) of the intensionally defined natural class trigger of the devoicing rule (the voiceless obstruents). This humble example reinforces the importance of *intensional* formulae as a component of the I-language approach in phonology, where it is less appreciated than in syntax. It does not appear to be the case that this kind of situation is what Hayes, Kornai and Mielke have in mind in positing the possibility of rules whose targets are not natural classes. In section 9, below, I address another apparent exception to natural class reference in rules, the case of apparent "L-shaped" classes.

3.3 Singleton sets and natural classes

Once we acknowledge that rules must refer to natural classes if we demand a useful theory of rules, a recurring confusion in the literature becomes clear. Suppose a rule deletes all nasals in some language, /m, n, ŋ/, word-finally. One might informally give the target of the rule as *m, n, ŋ*, or one might use a featural characterization of this set of segments, perhaps [+Nasal]. In either case, the rule is interpreted[4] as affecting any member of the set {m, n, ŋ} that occurs word-finally. There is a tendency, apparently due to a basic error of set-theoretic reasoning, to be inconsistent when the target class contains a single member.

In other words, a rule in another language that deletes, say, only /n/ word-finally, does not have the *segment* /n/ as its target, but rather it has the singleton set {n} as its target—and this singleton set can (and should) also be expressed intensionally in terms of a set of features, just as we did for the set {m, n, ŋ}. Below we formalize segments as sets of valued features, and natural classes of segments as intensionally characterized *sets of sets* of valued features.

I offer four pieces of evidence that this set-theoretic confusion, a failure to recognize the distinction between a singleton set like $\{x\}$ and the member of a singleton

[4]Most sources do not formalize the interpretation of phonological rule syntax—see Bale and Reiss (2018) for explicit interpretations of some simple rules.

set, x, is a potential source of confusion in the literature. My first argument is anecdotal: Chomsky feels obliged to make exactly this point on the blackboard in a lecture at MIT in 2019 (`https://youtu.be/GPHew_smDjY?t=327`): "the singleton set is different from its member." Anyone who has taught basic set theory has seen that students often mistakenly identify singleton sets with their members, however, a set must be a set, whereas the member of a set can be anything at all.

Second, consider this statement by Davenport and Hannahs (2010:91-92) "Natural classes may consist of any number of sounds, from two to many...". They seem to be under the impression that a set (or class) cannot contain a single member, and this reflects the false identification of a singleton set with its sole member.

A third sign of this error is the recurring claim that a natural class "is a class of segments that can be specified with fewer features than any individual member of the class" (Harms 1968:26). In the case of a singleton natural class, the same number of features is needed to specify the class and the sole member, as we will see below.

A fourth argument that singleton sets have been treated as not-quite-normal sets in the phonology literature is the fact that Wheeler (1972) specifically notes that he *will* treat them as normal sets in his discussion of the segment classes definable using the twenty-nine segments of Catalan: "Now the number of possible unordered sets (including sets with one member only) from a set of 29 different symbols is $2^{29} - 1$, that is, 536,870,91." In footnote, Wheeler thanks his sister Ruth for help with the math—we'll explain it more below. For now, note that the "-1" part reflects his intuition that the empty set of segments—another "weird" type of set, like the singletons—should not even be considered as a candidate natural class. As we will see below, my precise definition of which classes of segments are natural classes guarantees that the empty segment set is not a natural class—this is not left to Wheeler's (correct) intuition.[5]

Here's a statement that combines the error of entertaining the existence of rules that do not refer to natural classes with an error of set theoretic reasoning: "the languages of the world appear overwhelmingly to favor natural sets of sounds in their rules [and] this means that the languages of the world prefer to deal with sets of sounds that require few specified features for their identification rather than sets that require many" (Halle and Clements 1983:9). This statement is patently false: it implies that a rule that, say, assibilates /t/ to [s] before /i, as in Finnish, should be *less* common than a rule that assibilates all consonants before all vowels, since the class of all consonants is denoted by just [+CONSONANTAL], and the class of all vowels is denoted by just [−CONSONANTAL]. Similarly, Finnish raises only /e/ to [i] word-finally, despite the fact that characterizing the class containing just /e/ requires more features than characterizing, say, the set containing both /e/ and /æ/. Many, many rules target singleton natural classes, and those, in general, require more specified features to characterize than larger sets of segments. These confusions evaporate when we

[5]Perhaps my characterization of the error of reasoning is off—maybe the authors cited just think that it is okay to used a "mixed notation" (see below) for rules, using natural classes when the extension of the target has more than one member, and using a segment symbol in case the extension of the target has a single member. While the content of this mental state is distinct from the error of confusing a singleton set with its member, I think the two are related intimately enough to ignore the difference. If we assume that generalizations expressed in terms of features are real, things like the vowel harmony patterns of Turkish, then there is no advantage to also allowing for the expression of rules that refer to unanalyzable segments. Segments are just typographically convenient ways to refer to sets of features.

express rules in a manner that is precise with respect to set theoretic types.

4 Formalizing segments

In order to focus on feature logic, I abstract away from issues like redundancy in feature specification (e.g., that all nasals in English are voiced) and phonetic contingencies (e.g., that no vowel can be specified both +HIGH and +LOW). This abstraction is most easily achieved by considering a schematic universal phonology. I keep the math transparent by assuming a small inventory of universal features—it will be trivial to see how the results scale up for an inventory of more realistic size.

Let's suppose that UG_{toy} provides a set $\mathscr{F}$ containing just two features G and H, along with a set of values $V = \{+, -\}$.

(2) Primitives of UG_{toy}:

- Universal feature set: $\mathscr{F} = \{ \mathrm{G}, \mathrm{H} \}$
- Universal set of values: $V = \{+, -\}$

This UG yields, by the free combination of values and features (the cross product $V \times \mathscr{F}$), a set **F** of four valued features $\{ -\mathrm{G}, -\mathrm{H}, +\mathrm{G}, +\mathrm{H} \}$.

(3) Universal set of valued features: $\mathbf{F} = V \times \mathscr{F} = \{ -\mathrm{G}, -\mathrm{H}, +\mathrm{G}, +\mathrm{H} \}$

Abstracting away from other structures, such as timing slots, we then define a segment as a *consistent* subset of **F**. A segment ς is consistent if, for any F in $\mathscr{F}$, ς does not contain both +F and −F as members. Of course, **F** is not consistent.[6]

We can then define the universal segment inventory Σ as the set of consistent subsets of **F**, that is, the set of subsets of **F** that do not contains opposing valued features, like +G and −G. Each of these segment sets, then, contains −G, +G or neither, and −H, +H or neither. In other words, there are three possibilities for each feature in a segment: "+", "−" and "absent". With just two features, then, Σ has nine members.

I list the nine segments definable with UG_{toy} in (4) as members of the set Σ, numbered from 0 to 8.[7]

[6] In LP inconsistent sets can appear as the subtrahend (the change) of a rule built from the set subtraction operator: a rule can subtract, say, {+VOICED, −VOICED} from the members of the rule target class.

[7] The order of the nine segments from 0 to 8 corresponds to the ternary numbers from 0 to 22 using H for the *ones* column and G for the *threes* column. This scheme just provides a standard way of enumerating all the segments in a manner that is extensible for any number of features. This is an example of the Wolfram Code (https://en.wikipedia.org/wiki/Wolfram_code) that, in the simplest case, uses binary numbers to uniquely identify the 256 Elementary Cellular Automata (Wolfram 2002)—this is also useful for ordering the members of a power set in a consistent manner, as in fn. 8 below. Here I use ternary numbers to encode "+," "−" and "absent", just as the Wolfram Code uses ternary numbers for a three-color cellular automaton. See Unit 61 of Bale and Reiss (2018), where we introduce this system of enumeration and show that a schema for tone rules that yields 256 tone grammars with a tonal inventory containing just H and L explodes to over seven trillion possibilities just by the addition of a third tone, M.

(4) Universal segments in a two feature system:

$$\Sigma = \left\{ \begin{array}{ccccccccc} \{\,\} & \{-H\} & \{+H\} & \{-G\} & \{-G, -H\} & \{-G, +H\} & \{+G\} & \{+G, -H\} & \{+G, +H\} \\ \varsigma 0 & \varsigma 1 & \varsigma 2 & \varsigma 3 & \varsigma 4 & \varsigma 5 & \varsigma 6 & \varsigma 7 & \varsigma 8 \end{array} \right\}$$

Each member of Σ is denoted by an indexed ςi. Recall that each ςi must be consistent, but it need not be *complete*. For example, $\varsigma 7 = \{+G, -H\}$, and $\varsigma 3 = \{-G\}$. In other words, underspecification follows from the *absence* of a stipulation of completeness.

5 Formalizing natural classes

Most discussions of natural classes treat them on a language specific basis. We might say, for instance, that "In English {p, b, m} is a natural class but {p, m} is not, however in Cree {p, m} is a natural class, because the language has no /b/." I suggest that only the universal intensional notion of natural class is useful to understanding phonology. Natural class discussions can refer to surface or underlying inventories, but the derivational model assumed here also needs to take account of natural classes at intermediate levels. LP adopts the old idea (e.g., Harris 1984, Mester and Itô 1989, Poser 1993, Wiese 2000, McCarthy 2007, Samuels 2011) of treating feature-changing processes as a two-part sequence. In LP, one rule removes features (via set subtraction) and a subsequent rule inserts features (via unification), so there is no need for a primitive feature-changing operator (Bale et al. 2014, Reiss 2021). This approach also entails that turning, say, /d/ into /t/ requires an intermediate stage with an underspecified segment lacking a value for Voiced. In other words, devoicing of /d/ to /t/ occurs as /d/ ⇝ /D/ ⇝ /t/. In such a language, what is the inventory to be considered for determining natural classes? Under the approach I adopt here, this issue does not have to be considered—classes are not relativized to languages or derivational levels. For clarity, we can call a set of segments in a particular language, or at a derivational stage of a language, that are all and only the members of a universal natural class c_i, the *image* of c_i. Using our simple example above, the English set {p, b, m} and the Cree set {p, m} are images of the same (universal) natural class. We must distinguish (i) the intensional characterization of a (universal) natural class in terms of valued features; (ii) the (universal) extensional characterization of such a class in terms of (potential) segments; and (iii) the (extensional) image of the class in a particular language.

Under this perspective, some of the challenges facing Wheeler's 1972 discussion of natural classes become apparent: he based his analysis on the twenty-nine segments he posits for the phoneme inventory of Eastern Catalan. One issue is that each of Wheeler's "natural classes" is merely the Eastern Catalan image of a (universal) natural class—we don't want to formalize the notion of a class on the basis of the accidental image of the class that appears in a given language's inventory of lexical segments. A second issue is, as I argue below in section 11, that, in fact, Catalan contains segments that Wheeler did not consider, segments that differ from ones in Wheeler's inventory only with respect to underspecification of some features.

Given a universal inventory Σ of nine segments, there are $2^9 = 512$ possible sets of segments. For each of these *sets*, we have to see whether each *segment*, $\varsigma 0$, …, $\varsigma 8$, is in the set. In other words, we ask "Is $\varsigma 0$ in this set?" and "Is $\varsigma 1$ in this set?," and so on. We need to ask nine Y/N questions so there are $2^9 = 512$ possible sets of answers corresponding to 512 possible sets of segments. So, the set of sets of segments definable by UG_{toy}, is the *power set* of the set of segments—and if a set has cardinality n, its power set has cardinality 2^n, as in our example.[8]

We want to know which of the 512 subsets of Σ, which members of the powerset, are *natural* classes. This is an old question: "any theory which can discount the vast majority of these classes as 'un-natural' has considerable power. So we want not merely a powerful theory but a correct theory of exactly the right power" (Wheeler 1972). In order to answer the question we need a definition of natural class.

We define *natural* classes, following a long tradition, via a conjunction of consistent valued features.

(5) **Definition of natural class:** A subset Q of the universal segment inventory Σ is a natural class iff there exists a consistent set of valued features $\mathscr{C}$ such that
$Q = \{q \mid q \supseteq \mathscr{C}\}$

Since each q is a segment, each q must be consistent. If $\mathscr{C}$ were not consistent, there could be no segment q that is a superset of $\mathscr{C}$—any superset of an inconsistent set will be inconsistent. For concreteness, the natural class of high, round vowels, can be expressed as follows:

(6) High, round vowels: $Q = \{q \mid q \supseteq \{+\textsc{High}, +\textsc{Round}\}\}$

This set-builder notation makes it clear that a natural class is type-theoretically a set of sets of valued features, an intensionally characterized partial description of a set of elements (segments) that all subsume a particular information structure (a set of features). Because this notation can be unwieldy, LP introduces a useful abbreviation. We reserve square brackets for natural class representations, allowing us to rewrite (6) as (7).

(7) High, round vowels in square bracket notation:

$$\{q \mid q \supseteq \{+\textsc{High}, +\textsc{Round}\}\} = \begin{bmatrix} +\textsc{High} \\ +\textsc{Round} \end{bmatrix}$$

The set builder notation helps us uncover another common confusion in natural class reasoning, namely the inverse relation between between the number of valued features used to intensionally define a partial description, a set of representations, and the number of members in the set. I refer to this as the Rudy Valentino Principle (RVP), after the "tall, dark and handsome" silent film actor of the 1920's, which is more memorable than a name like the *inversion of extension-intension subsumption.* Clearly, a superset of features yields a partial description with a subset of entities.

[8]We can use the Wolfram code here, too, with powers of 2: the set of all segments corresponds to the answer "Yes, this segment is in the current subset" for each of the nine segments. Converting "Yes" to "1" yields the binary number 111111111 which is subset 511 of Σ, namely Σ itself; and the empty set of segments corresponds to a "No" for each segment, yielding the binary number 000000000=0, which is subset 0, the first subset of Σ. Counting like computer scientists, from 0 to 511, thus yields $2^9 = 512$ sets of segments, each a subset of Σ.

(8) Rudy Valentino Principle (RVP):

> If there is an intensional description of set A which subsumes (is more general than) the intensional description of a set B, then any element of the extension of B is in the extension of A (that is, B is a subset of A)

To illustrate, let A be the set of tall men, A={x| x is tall}, and let B be the set of tall, dark, handsome men, B ={x| x is tall, dark and handsome}. The set of properties in the intensional characterization of this A is a subset of the set in the intensional characterization of B: {tall} ⊆ {tall, dark, handsome}. However, the relation is inverted if we look at the elements in the extensions of the two sets: $B \subseteq A$, or using our square bracket notation, [tall, dark, handsome] ⊆ [tall].

In Hale and Reiss (2003) we argue that failure to appreciate this point of logic (which we had not yet named) has led to a completely inverted interpretation of the Subset Principle of acquisition in phonology. We apply RVP directly to phonological natural classes below.

6 Traditional notation is not type-consistent

Note now that traditional rule notations are not type-consistent. They do not reflect the fact the target and trigger of a rule express natural classes, but that the change effected by the rule refers to a change in content of segments, not to a class of segments. In LP, the target and environments are expressed with square brackets, whereas the change is expressed with normal set theoretic brackets—the change is a set of valued features.[9]

Rules are sometimes informally expressed extensionally in terms of segments as in (9).

(9) Extensional formulation:

o, e	→	u, i	/ __	m, n, ŋ
(Target)		(Change)		(Environment)

However, in line with the discussion of singleton segments sets above in section 3.3, Chomsky and Halle (1965) make it clear that "phonological rules [must] be stated completely in terms of features, rather than in terms of phonemes or in a mixed notation," as in the traditional generative format of (10).

(10) Traditional formulation:

[−Low]	→	[+High]	/ __	[+Nasal]
(Target)		(Change)		(Environment)

LP goes further in formalization by distinguishing the set theoretic difference between sets of segments corresponding to natural classes (in square brackets) from the set of features that express a change (in curly brackets), as in (11).

[9] Every segment is a set of valued features, but not every set of valued features is a segment.

(11) Revised LP formulation:

$$\begin{array}{lllll} \big[-\textsc{Low}\big] & \rightarrow & \big\{+\textsc{High}\big\} & /\_\_ & \big[+\textsc{Nasal}\big] \\ \text{(Target)} & & \text{(Change)} & & \text{(Environment)} \end{array}$$

In brief, it is crucial to pay attention to brackets in order to be set-theoretically consistent. Natural classes can be denoted intensionally using square brackets or the set builder notation, but also extensionally, by listing a set of segments, such as {p, b, m}.

7 Circle notation

Before returning to the question of which subsets of Σ are natural classes, we introduce one more piece of notation. When we work with a model of UG that has more than just two features, the list of features used to characterize a natural class can become quite long. We can always select an arbitrary symbol to stand in for a set of features, but in order to make convenient use of the square bracket notation we need a way to refer to a *list* of features without denoting them as members of a set. We thus introduce the circle notation. A circled symbol like $\mathscr{C}$ denotes the list of features contained in the set $\mathscr{C}$. We then use square brackets and a circled symbol to denote a natural class in a compact manner.

(12) Natural class notation:

$$\{q \mid q \supseteq \mathscr{C}\} = [\mathscr{C}]$$

Returning to the example of high round vowels in (6), above, suppose we want to refer to this class in a language that has only the following high round vowels in its inventory /y, ʏ, u, ʊ/. We can use a symbol like “Y” to denote the set {+High, +Round}, even though no actual segment /Y/ occurs in the language. Thus, the relevant natural class will be Y. This circled symbol notation will be handy below—just think of it as erasing one level of set brackets—so /t/ = {t}.[10]

8 The natural classes of UG$_{toy}$

Now note that the set of all consistent subsets of **F** is exactly the nine sets listed as *segments* in (4). Each one of these sets has one of −G and +G, but not both; or neither. The same holds true for feature H. Any other subset of **F** is not consistent.

If we set the value of $\mathscr{C}$ in (12) to the value of each of the segments in (4), we get the following natural classes.

(13) Natural Classes defined by conjunctions of valued features:

[10]Forward slash “/” is used purely typographically to identify segment symbols where needed.

	Natural class	Extension	Set builder notation
0	[ ]	$\{\varsigma 0, \ldots, \varsigma 8\}$	$\{q \mid q \supseteq \{\ \}\}$
1	[−G]	$\{\varsigma 1, \varsigma 4, \varsigma 7\}$	$\{q \mid q \supseteq \{-\mathrm{G}\}\}$
2	[+G]	$\{\varsigma 2, \varsigma 5, \varsigma 8\}$	$\{q \mid q \supseteq \{+\mathrm{G}\}\}$
3	[−F]	$\{\varsigma 3, \varsigma 4, \varsigma 5\}$	$\{q \mid q \supseteq \{-\mathrm{F}\}\}$
4	[−F, −G]	$\{\varsigma 4\}$	$\{q \mid q \supseteq \{-\mathrm{F}, -\mathrm{G}\}\}$
5	[−F, +G]	$\{\varsigma 5\}$	$\{q \mid q \supseteq \{-\mathrm{F}, +\mathrm{G}\}\}$
6	[+F]	$\{\varsigma 6, \varsigma 7, \varsigma 8\}$	$\{q \mid q \supseteq \{+\mathrm{F}\}\}$
7	[+F, −G]	$\{\varsigma 7\}$	$\{q \mid q \supseteq \{+\mathrm{F}, -\mathrm{G}\}\}$
8	[+F, +G]	$\{\varsigma 8\}$	$\{q \mid q \supseteq \{+\mathrm{F}, +\mathrm{G}\}\}$

Let's illustrate with natural class 2. In this case, $\mathscr{C}$ has the value $\varsigma 2$. Using the circle notation, then, we derive this expression: $\{q \mid q \supseteq \{\varsigma 2\}\} = \{q \mid q \supseteq \varsigma 2\}$. And this is equivalent to $\{q \mid q \supseteq \{+\mathrm{G}\}\}$. In other words, natural class 2 is the set of segments that are supersets of $\varsigma 2$, including of course, $\varsigma 2$ itself. Parallel reasoning holds for each ς. Nothing else is a natural class.

Several result are immediately apparent. First, we see that of the 512 set of segments definable with UG_{toy} only nine of them are natural classes. Next, we see that only a fully specified segment corresponds to a singleton natural class. For example, natural class 8 is [+F, +G] = [$\varsigma 8$] = $\{\varsigma 8\}$ contains only the segment {+F, +G} = $\varsigma 8$. In contrast, no underspecified segment, say, $\varsigma 6$, will constitute a natural class by itself (in terms of the universal segment inventory Σ, although it might reflect the image of a natural class in a particular language).

The generalization of this result, which is just a reflection of the Rudy Valentino Principle, is called *specificity* in recent LP work (e.g., Gorman and Reiss 2025b), a notion that was understood already by Lees (1961), but which has been underappreciated.

(14) **Specificity:** Let f, g be segments such that $f \subset g$. Then there is no natural class containing f but excluding g.

We now see even more clearly why Harms (1968:26) and others are wrong in the recurring claim mentioned above that a natural class "is a class of segments that can be specified with fewer features than any individual member of the class." In the case of a singleton natural class like class 8, the same number of features is needed to specify the class and the sole member.

To return to our concrete example of {t, D, d}, it is now apparent that {t} is the natural class [t] and {d} is the natural class [d].[11] However, the class [D] necessarily contains {t, D, d}. Similarly, neither {t, D} nor {D, d} are natural classes. If a natural class contains /D/, it must contain every segment that is a superset of /D/.[12] As we saw, a case parallel to a class containing just /D/ in our toy system is the set of segments $\{\varsigma 6\}$ containing just $\varsigma 6$. Note that this class does *not* appear in (13). This set cannot be a (universal) natural class, since other segments ($\varsigma 7$ and $\varsigma 8$) are supersets of $\varsigma 6$.

[11] Note that it is incorrect to write these as [t] and [d], without circle notation.

[12] To reiterate the discussion from section 2, these results do not hold in a system that allows reference to, say, "not +VOICED," but our UG_{toy} does not provide for that possibility.

Note that natural class 0 is the universal natural class containing all segments, since every segment is a superset of the empty set of valued features: for each *i*, $\varsigma i \supseteq \varsigma 0$. This means that we can use "[ ]", which is equivalent to [$\varsigma 0$], to denote "any segment" and we don't need a feature ±Segment (Bale et al. 2019)—our definition of natural classes plays a role in determining the scope of what can be expressed in rules.

A further implication is that the empty set of *segments* is *not* a natural class (as Wheeler intuited). This follows from the definition of natural class in (5). Whatever consistent set is chosen for $\mathscr{C}$ in the definition, there will always be at least the segment $\mathscr{C}$ in the natural class. (And every segment that is a superset of $\mathscr{C}$ will also be there. So if $\mathscr{C}$ is $\varsigma 0$, every other segment will be present in the class, as well.) In other words, $\mathscr{C}$ will always be a member of [$\mathscr{C}$].[13] It then follows that we need boundary symbols like "#". We can't write, say, "after no segment" to mean "at the beginning of a domain" without a boundary symbol, because there is no way to express the empty class of segments. Again, the precise definition of natural classes has implications for what rules can express and, in this case, forces us to include boundary symbols in our ontology.

We now have a partial theory of what can and cannot be the targets and triggers of rules, a theory of the "scope and limits" of phonology. LP research consists in large part in showing how paying attention to specificity allows our austere model to expand the scope of phonology beyond what has traditionally been believed to be the case by offering narrow phonological account of phenomena that have been wrongly supposed to require lexical or morphosyntactic conditioning or exception features (e.g., Gorman and Reiss 2025a). Specificity may also tell us when an apparent pattern cannot be formulated as a single rule.

9 The illusion of "L-shaped natural classes"

In order to show concretely how underspecfication interacts with natural class logic, and to show that putative cases of rules referring to non-natural classes can be reanalyzed in a manner consistent with *specificity*, consider the case of what Mielke (2008:131) calls an "L-shaped" class.

Mielke states that in a certain environment three of the four high vowels of the Bantu language Kinyamwezi become glides rather than assimilating to a following vowel, as the other vowels do. The three that become glides are /i, u, ʊ/, whereas /ɪ/ does not undergo this process. Mielke presents the vowel chart in (15).

(15) Example of Mielke's "L-shaped class" as rule target:

i		u
ɪ		ʊ
e		o
	a	

[13]Thanks to Rim Dabbous for making me see this.

Obviously, no theory of natural classes based on innate features will treat /i, u, ʊ/ as a natural class to the exclusion of /ɪ/. Since the "relation between a phonemic system and the phonetic record ...is remote and complex" (Chomsky 1964:38), we are not constrained to posit as the underlying representation for the vowel that surfaces only as [ɪ], a fully specified segment. Instead, I propose the partial representation of the Kinyamwezi vowels in (16).

(16) Partial feature specifications for Kinyamwezi vowels:

	/i/	/I/:[ɪ]	/u/	/ʊ/	/e/	/o/	/a/
High	+	+	+	+	−	−	−
Low	−		−	−	−	−	+
Atr	+	−	+	−	+	+	−

The target of the rule that turns three of the surface high vowels into glides can be characterized as the set of vowels specified +High and −Low. Crucially, in this analysis, Kinyamwezi has no lexical /ɪ/.

(17) The apparently L-shaped class: $\begin{bmatrix}+\textsc{High}\\ -\textsc{Low}\end{bmatrix}$

The image of a natural class in the lexicon of a language may be L-shaped, but in a rule, there are no L-shaped classe, since all classes are defined intensionally using conjunctions of valued features from **F**. The vowel that surfaces as [ɪ] is underlyingly /I/, by hypothesis, without a specification for Low. This vowel is not a superset of the features used to define the target of the gliding rule.

(18) $\begin{Bmatrix}+\textsc{High}\\ -\textsc{Atr}\end{Bmatrix} \not\supset \begin{Bmatrix}+\textsc{High}\\ -\textsc{Low}\end{Bmatrix}$

While the choice of underspecifying /I/ for Low looks arbitrary, it is consistent with the fundamental principle that the phonological behavior of a segment follows from its phonological make-up in the context of a set of rules. Before throwing out the baby of innate features, we should fully appreciate the potential of the specificity-infused bathwater.[14] In any case, the narrower point to take away is that L-shaped classes may only be L-shaped in extension with respect to the segment inventory of a particular language. In actual grammers, rules make reference to natural classes defined intensionally by conjunctions of features. More concretely, any class that contains /i, u, ʊ/ *must* include /ɪ/ as well, but it may be the case that the latter does not

[14] One might be tempted to say that this system can be "abused," for example, by allowing for a language with the same four surface high vowels as Kinyamwezi to treat, say, /i/ and /ʊ/ as a natural class by leaving both [ɪ] and [u] underlyingly unspecified for Low. However, this is exactly what a feature system is supposed to do, to capture the behavior of sets of segments. Of course, the model will allow for some unattested patterns, but there is no reasonable way to restrict a phonological model that allows underspecification from ruling out such cases. As our discussion of combinatorics below shows, this is an unavoidable property of any combinatoric system which will have to be weighed against other factors. We do not expect our model of UG to cover only the attested languages, any more than we expect chemistry to cover only attested substances—after all, chemists *invented* plastics (Reiss 2022). The "diachonic filter" (Hale 2003) and not UG, is in large part responsible for what generable patterns are actually attested.

exist in a given language's lexicon, in other words, it is not in the image of that class in that language. A large part of the LP literature consists in identifying such cases of absolute neutralization, cases where underlying segments distinguished from others only by virtue of underspecification do not have any unique surface form of their own. A classic case is the Turkish /t, D, d/ constrast discussed by Inkelas (1995), Inkelas et al. (1997). Gorman and Reiss (2025a) contains several other cases, and Gorman (to appear) is a particularly clear and concise example.

10 Rule operators in Logical Phonology

LP deconstructs the traditional arrow "→" of phonological rules into three basic operations. Further research exploring phonemomena such as syllabification and stress computation will surely expand the LP arsenal, but for now, we restrict discussion to two intrasegmental operations— one that removes features from a segment and one that adds features to a segment—and two that affect whole segments—one that inserts whole segments and one that deletes whole segments.

10.1 Intrasegmental operators: subtraction and unification

The two operations that subtract features from a segment or add features to a segment can be discussed by treating segments just as sets of valued features. The first operation from which a rule can be constructed is set subtraction, also called set *difference*. I use the set subtraction operator with the standard definition from mathematics: If A and B are sets, then $A \smallsetminus B = C$, where $C = \{c : c \in A \text{ and } c \notin B\}$. That is, the members of $A \smallsetminus B$ are all the elements of A except for those that are in B. Here are some examples:

(19) Examples of set subtraction:

- $\{x, y\} \smallsetminus \{x, z\} = \{y\}$
- {+Voic, −Son, +Cor} $\smallsetminus$ {+Voic, +Son} = {−Son, +Cor}

In each of these examples, note that the subtrahend, the set that is being subtracted (the one on the right of the '$\smallsetminus$' symbol), can contain elements that are not present in the minuend, the set on the left of the operator. Those elements are irrelevant to the outcome of the computation. Set subtraction rules model so-called delinking in autosegmental models, for example, rules that remove associations between timing slots and featural content. Such rules make segments less specified, and sometimes this derived underspecification persists to the surface, as in the case of Arbore debuccalization discussed by McCarthy (2008), where glottalized stops lose all place features in coda position and neutralize to placeless glottal stops. Subtraction rules also model the first part of a feature changing process, for example, turning an underlying /d/ into /D/ by subtracting the set {+Voiced}.

(20) A subtraction rule: [ⓓ] $\smallsetminus$ {+Voiced}

This use of a segment deletion rule is illustrated below.

The second operation is a set theoretic version of unification, which is widely used in syntax (e.g., Shieber 1986). LP unification is closely related to standard set union. *Union* combines the members of sets into a single set: If A and B are sets, then $A \cup B = C$, where $C = \{c : c \in A \text{ or } c \in B\}$. Union is called a "total operation" since the union of any two sets always yields a set. Instead of union, we use a version of unification, which is typically used to combine expressions that are more complex than sets. Unification is useful in LP because the members of segment sets come from **F**—they are ordered pairs of values ('+' or '−') combined with features. Unification $\sqcup$ requires the phonology to look into the valued features that are the elements of feature sets in order to evaluate consistency. If A and B do not contain opposites, then $A \sqcup B = A \cup B$. However, if A and B do contain conflicting values, that is if $A \cup B$ is not consistent, then $A \sqcup B$ is *undefined*. This is referred to as unification *failure*. When unification failure occurs in the application of unification in a phonological rule, LP rule semantics treats this as a case of vacuous application.

Consider the following rule and its yield when applied to various inputs.

(21) Rule: $\begin{bmatrix} -\text{Back} \\ -\text{Round} \end{bmatrix} \sqcup \{-\text{High}\}$

a. "Feature addition": $\begin{Bmatrix} -\text{Back} \\ -\text{Round} \end{Bmatrix} \sqcup \{-\text{High}\} \rightsquigarrow \begin{Bmatrix} -\text{Back} \\ -\text{Round} \\ -\text{High} \end{Bmatrix}$

b. "Vacuous application": $\begin{Bmatrix} -\text{Back} \\ -\text{Round} \\ -\text{High} \end{Bmatrix} \sqcup \{-\text{High}\} \rightsquigarrow \begin{Bmatrix} -\text{Back} \\ -\text{Round} \\ -\text{High} \end{Bmatrix}$

c. "Unification failure": $\begin{Bmatrix} -\text{Back} \\ -\text{Round} \\ \underline{+\text{High}} \end{Bmatrix} \sqcup \{\underline{-\text{High}}\} \rightsquigarrow \begin{Bmatrix} -\text{Back} \\ -\text{Round} \\ \underline{+\text{High}} \end{Bmatrix}$

Case (a) is feature-filling, since the valued feature −High is added to the input segment set. Case (b) shows vacuous application of the rule, because the input segment is already specified −High. Case (c) is also vacuous application, because set union would yield an inconsistent set, so unification fails. This use of unification allows LP to *appear* to target only relatively underspecified segments, but actually doing so violates specificity: intensionally, specificity is always obeyed. The two kinds of vacuous application model the effects of purely feature-filling processes, as illustrated in various works (Bale et al. 2014, Bale and Reiss 2018, Reiss 2021, Gorman and Reiss 2025a), without violating the theory of natural classes developed above.

10.2 Feature-changing processes

Following the suggestions of Poser (1982), Inkelas and Cho (1993), and Siptár and Törkenczy (2000) LP models feature-changing processes as subtraction followed by unification, as in the case of Russian final devoicing of obstruents exemplified by d ⇝ D ⇝ t.

(22) Russian final devoicing:

	nom.sg.	gen.sg.	
a.	$\widehat{\text{ts}}\text{v}^{j}\text{et}$	$\widehat{\text{ts}}\text{v}^{j}\text{eta}$	'color'
b.	prut	pruda	'pond'

(23) Consonant features:

	/t/	/d/	/D/
Voice	−	+	
Sonorant	−	−	−

The first rule subtracts the set {+Voiced} from any word-final obstruent—this rule will of course apply vacuously to voiceless obstruents.

(24) Part 1 (addition): $\left[-\textsc{Sonorant}\right] \setminus \left\{+\textsc{Voice}\right\} \;/\; \_\_\ \%$

Rule (24) maps any word-final segment-set that is a superset of {−Sonorant} to that same segment-set minus {+Voice}. The next rule unifies any obstruent with the set {+Voiced}.

(25) Part 2 (removal): $\left[-\textsc{Sonorant}\right] \sqcup \left\{-\textsc{Voice}\right\}$

Depending on the specifications on input target segments, the yield is as in (26).

(26) Yield of (25):

a.	$/\text{t}/ \sqcup \{-\textsc{Voice}\} \rightsquigarrow /\text{t}/$	(vacuous unification)
b.	$/\text{d}/ \sqcup \{-\textsc{Voice}\} \rightsquigarrow /\text{d}/$	(unification failure)
c.	$/\text{D}/ \sqcup \{-\textsc{Voice}\} \rightsquigarrow /\text{t}/$	(feature filling)

As above, the rule merely appears to target /D/ to exclusion of /t/ or /d/. Unification allows us to state the rule without a context, since segments like /D/ which will get filled in only appear as the result of rule

10.3 Segment Deletion and Insertion

In LP, a third operator "$\mapsto$" is used along with a null segment symbol "ϵ" to implement insertion and deletion of entire segments. Note that LP does *not* model segment deletion via subtraction. Subtraction removes features, not segments, and segmental underspecification is representationally distinct from segmental absence. An empty set is not nothing—it is a set.

For subtraction and unification we focussed on segments as sets of valued features. Once we get to insertion and deletion of full segments, we have to enhance our representations to include association of the sets of valued features to timing slots. Segment insertion involves insertion of a set of valued features and an associated timing slot. Segment deletion involves the deletion of a slot and its set of valued features.

In insertion rules, the X is linked to a feature bundle and a specific segment is inserted, so the structural change is an X-slot associated to a particular set of features (in curly brackets). For example, an insertion rule must insert a particular segment, say the vowel [i]. In deletion rules, however, the X is linked to a natural class (expressed with square brackets) and any segment matching the natural class is deleted. So, a deletion rule can delete, say, any member of the set of high vowels.

(27) Insertion schema:

$$\epsilon \;\mapsto\; \overset{\substack{\mathrm{X}\\ |}}{\{\ldots\}} \;/\; [\ldots] \;\_\_\; [\ldots]$$

(28) Deletion schema:

$$\overset{\substack{\mathrm{X}\\ |}}{[\ldots]} \;\mapsto\; \epsilon \;/\; [\ldots] \;\_\_\; [\ldots]$$

Because of the reference to natural classes, targeting in deletion rules conforms to Specificity. In unification rules, this Specificity effect is concealed by vacuous application and unification failure, but there is no such escape hatch for deletion rules. That is, a unification rule of the form [Ⓓ] $\sqcup$ {+Voiced} has an observable effect only on /D/, and not /t/ or /d/. In contrast, a deletion rule targeting [Ⓓ] also targets /t, D, d/ intensionally, but necessarily deletes all three, as we discuss further in the next section.

11 Delete the rich in Catalan

As mentioned above, specificity is just the application of RVP to phonological representations. Following up on this logic we derive a result called Delete the Rich.

(29) Delete The Rich (Reiss 2025, Gorman and Reiss 2025b): If some—but not all—X's delete in some phonological context, the X's that delete must be **more richly specified** than those which do not.

In other words, suppose a language has some lexical items that sometimes surface with the segment [ϕ], but that ϕ deletes in a certain context T: ϕ alternates with the null segment ϵ. Suppose the language has other lexical items that also surface with ϕ, but in these lexical items, ϕ does not delete in context T. We can model the two kinds of ϕ, that is, ϕ_1, the one that alternates with ϵ, and ϕ_2, the one that does not alternate, as being in a subset relation due to underspecification (i.e., $\phi_1 \supseteq \phi_2$). If we do that, it is necessarily the case that the *non*-alternating segment ϕ_2 be relatively *less* specified than the alternating ϕ_1. Continuing the notational convention of using capital letters for underpsecification, we might replace ϕ_2 by /Φ/ and let ϕ_1 be just /ϕ/. Let's think in terms of specificity: if $\Phi \subset \phi$, then any natural class containing Φ must contain ϕ. However, /ϕ/ can constitute a natural class that excludes /Φ/.

In honor of Wheeler's early paper on natural classes based on the segments of Catalan, let's illustrate with an example from this language, using data confirmed by Wheeler himself (p.c.). Catalan has a rule that deletes word-final *r*, as in the masculine singular *kla*. In contrast the *r* of *pur* is not deleted word-finally (and is typically realized as a flap or tap).

(30) Regular and 'exceptional' final /r/ in Catalan:

Masc Sg	Fem Sg	Lexical Form	Gloss
kla	klarə	/klar-/	plain
pur	purə	/puR-/	pure

Contrary to intuition, it must be the less specified final segment of “pure”, denoted /R/, that fails to delete, because it is possible to specify the singleton target class [ⓡ], but the class [Ⓡ] necessarily contains both /r/ and /R/. One solution that appears to work when more data is considered is to treat /R/ as underspecified for Lateral, with /r/ fully specified. Parallel reasoning yields also a deleting and non-deleting pair of nasals, /n/ and /N/.

(31) Regular and ‘exceptional’ final /n/ in Catalan:

Masc Sg	Masc Pl	Lexical Form	Gloss
bo	bons	/bon-/	good
kon	kons	/koN-/	cone

Since both /r/ and /n/ delete word-finally, we can posit a single rule that effects this deletion. It targets both the fully specified segments, but not /R/ and /N/. By hypothesis, the latter two are unspecified for Lateral whereas /r/ and /n/ are specified −Lateral. The target of the rule, then will be defined by the set of features common to /r/ and /n/, the non-lateral coronal sonorants, for which I use the symbol $\mathscr{R}$.

(32) $\mathscr{R} = \{-\text{Syllabic}, +\text{Sonorant}, +\text{Coronal}, -\text{Lateral}, \ldots\}$

The target of the deletion rule is thus the natural class built from the valued features of $\mathscr{R}$, by hypothesis a list containing specification for every feature except Nasal.

(33) Word-final $\mathscr{R}$-deletion—target class is {r, n}:

$$\begin{array}{c} \text{X} \\ | \\ [\mathscr{R}] \end{array} \mapsto \epsilon \;/\; \_\_ \;\%$$

Catalan has another deletion rule affecting only /r/, but not /R, n, N/. Before coda *-s*, only /r/ is deleted, as in *klas*.[15]

(34) Deletion before coda *-s* of /r/ in Catalan: :

Masc Sg	Fem Sg	Masc Pl	Lexical Form	Gloss
kla	klarə	klas	/klar-/	plain
pur	purə	purs	/puR-/	pure
bo	bonə	bons	/bon-/	good
kon		kons	/koN-/	cone

The target of this segment deletion rule is [ⓡ], the singleton natural class containing just /r/.

[15] We know that there is not one single rule deleting /r/ in coda, because of forms like [for.tes] ‘strong’ Masc Pl.

(35) /r/-deletion before /s/—target class is {r}:

$$\begin{array}{c} \mathrm{X} \\ | \\ [\text{ⓡ}] \end{array} \mapsto \epsilon \ / \ \_\_ \ [\text{ⓢ}]$$

The environment is defined as "before *s*", which of course, must be represented in terms of the singleton natural class, shown here using the circled *s* as an abbreviation for a list of features that define a "partial description" that, in this case, is specified for all features.

Since, /R/ and /N/ neutralize fully with /r/ and /n/, respectively, an LP analysis requires a context-free feature-filling unification rule that targets /R/ and /N/ and fills in the valued feature −LATERAL. However, by specificity, it is not possible to target /N/ without targeting /n/, since the former is a subset of the latter, and the same reasoning holds for /R/ and /n/. It is also not possible to target /R/ without targetting /l/, assuming that they are identical aside from the specification +LATERAL in /l/, where /R/ has no specification. The LP solution is to target *all* coronal sonorants, including the +LATERAL /l/. I use the symbol $\mathscr{N}$ for this class. In brief, the rule works just like the illustration in (21).

(36) Context-free fill in rule: $[\mathscr{N}] \sqcup \{-\text{LATERAL}\}$

a. Unification is feature-filling: /R/, /N/ ⊔ {−LATERAL} yields [r], [n], respectively

b. Unification is vacuous: /r/, /n/ ⊔ {−LATERAL} yields [r], [n], respectively

c. Unification fails: /l/ ⊔ {−LATERAL} yields [l] (vacuous by LP rule semantics)

The unification rule applies non-vacuously when features can be filled in, as in the mapping of /R, N/ to [r, n], respectively. When union is vacuous, the unification rule itself applies vacuously, as in the case of /r/ and /n/. In the case of /l/, unification fails, and LP rule semantics treats this, too, as vacuous unification.

To review a bit, note that /r, R, n, N, l/ are actual segments in the Catalan lexicon. The symbols $\mathscr{R}$ and $\mathscr{N}$ are potential symbols that happen not to occur in the Catalan lexicon. Catalan rules use circled versions of the symbols "r, $\mathscr{R}$" and "$\mathscr{N}$" to denote natural classes. Only the first, "r," yields a singleton natural class, with a singleton image in Catalan. The symbol "$\mathscr{R}$" yields a natural class with the image {r, n} in Catalan. Finally, the symbol "$\mathscr{N}$" yields a natural class with the image {r, R, n, N, l} in Catalan.

Following the logic of specificity carefully lead us to discover the limitations of our intuitions concerning deletion. Since DELETE THE RICH follows from basic set theory including RVP, and not from any language specific principles, we can perhaps treat it as an example of a "third factor" effect (Chomsky 2005). Delete the Rich is not directly encoded in the language faculty, so it is not part of innate UG, Chomsky's first factor of language design. If, as suggested here, Delete the Rich is a necessary property of every phonological system, then language specific experience has no bearing on the I-languages attained with regard to Delete the Rich. So, the second factor, experience, is irrelevant. The only remaining option for explaining Delete the Rich is that of the third factor, " Principles not specific to the faculty of language, " and that is how we have treated this result, as following from mathematical logic.

Note that the LP approach has allowed us to reduce the "exceptional" behavior of some *r*'s to pure phonology. With the right computational system and the right representations, Catalan has no lexical exceptions. Crucially, no enhancement to the LP model is required to achieve such a result. I analyzed this aspect of Catalan with two segment deletion rules and one unification rule.

12 Discussion

Returning to our questions in (1), we see that given a UG with n features in $\mathscr{F}$, the number of valued features in **F** is 2^n, and the number of possible segments is 3^n. The number of natural classes is equal to the number of segments, since each class is defined intensionally as the set of supersets of some segment. If n is just 4, this yields $3^4 = 81$ segments, and so, 81 natural classes. Of course the number of classes, sets of segments, natural and unnatural, is 2^{81}, or about 2.4×10^{24}. A reduction from this astronomical number of classes (according to Wolfram Alpha, this is more than the number of grains of sand on earth) to a mere 81 *natural* classes is a nice result. If n is instead 20, still a fairly conservative number of features to posit for UG, this yields 3^{20}, over a billion, segments, and thus over a billion natural classes. This large number is still significantly smaller than the total number of classes of segments, 2^{10^9}, that is, 2 to the *billionth* power. The latter is significantly more than the 2^{285} number of particles in the universe, a magnitude which Gallistel and King (2009) call "essentially infinite."

The takeaway point of all this combinatorics is that our precise notion of natural classes restricts by orders of magnitude the set of possible rules by requiring that the target and trigger classes of rules be natural classes. It is hard to imagine any restrictions whatsoever in an approach like that of Mielke (2008) or Archangeli and Pulleyblank (2015) that adopts neither innate features nor the requirement that rules be formulated in terms of natural classes of these features.

Two more simple combinatoric points can be made with respect to our original questions in (1). First, if there is no underspecification, the number of segments reduces from 3^n to 2^n, since "absence" is not a choice for features in a segment. Powers of 2 obviously grow more slowly than powers of 3, but 2^n still yields a lot of possibility from a small UG. Finally, we can also think of the number of classes of segments, the subsets of a universal segment set Σ, as the set of possible segment inventories that a language can have. In other words, a UG with twenty features, the values + and −, along with underspecification, can define $2^{3^{20}}$, that is, more than 2^{10^9}, distinct languages (qua inventories). While this is not evidence for the existence of an innate UG, it does make for a plausibility argument: the more diversity we get via combinatoric explosion, the less we have to attribute to the genome in order to account for the "welter of descriptive complexity" (Chomsky 2000) observed in attested languages. In other words, this is a corollary to Chomsky's (2007) point about the idea of a UG encoded in the genome and instantiated in the brain that the "less attributed to genetic information … the more feasible the study of its evolution" and its neural implementation.

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